

UNIVERSITY OF ZAGREB
FACULTY OF ELECTRICAL ENGINEERING AND COMPUTING

MASTER THESIS No.

**DEEP LEARNING-BASED ANIMAL
RE-IDENTIFICATION FOR NON-INVASIVE
WILDLIFE MONITORING AND
CONSERVATION**

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Zagreb, June, 2025

SVEUČILIŠTE U ZAGREBU
FAKULTET ELEKTROTEHNIKE I RAČUNARSTVA

DIPLOMSKI RAD br.

**DUBOKO UČENJE ZA PONOVRNU
IDENTIFIKACIJU ŽIVOTINJA U
NEINVAZIVNOM PRAĆENJU DIVLJIH VRSTA I
OČUVANJU PRIRODE**

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Zagreb, lipanj, 2025.

The master thesis assignment text in English goes here.

Ovdje dolazi tekst zadatka diplomskog rada na hrvatskom jeziku.

Thanks for the tea...

Contents

1	Introduction	3
2	Related Work	5
2.1	Traditional Computer Vision Approaches	5
2.2	Species-Specific Trait-Based Re-Identification	6
2.3	Deep Learning-Based Methods	6
2.4	Hybrid Approaches with Learned Local Features	7
2.5	Key Challenges and Limitations	7
3	Dataset and Preprocessing	8
3.1	Public Benchmark Dataset: WildlifeReID-10k	8
3.1.1	Dataset Overview	8
3.2	Original Dataset Collection: Medvednica	8
3.2.1	Dataset Collection Methodology	9
3.2.2	Automatic Animal Detection and Classification	9
3.2.3	Preprocessing Pipeline Overview	10
4	Feature Extraction	11
4.1	Local Feature Detection	11
4.2	Descriptors	12
4.3	DISK: Learning Local Features With Policy Gradient	13
4.4	KeyNet, AffNet, and HardNet	13
4.4.1	KeyNet	13
4.4.2	AffNet	13
4.4.3	HardNet	14
4.5	Storage Format and Management	14

5	Feature Aggregation and Representation	16
5.1	Training PCA and GMM models	16
5.2	Fisher Vector Computation	17
5.3	Parameter Choices and Rationale	18
6	Geometric Verification	20
6.1	Theoretical Foundation	20
6.2	Establishing Initial Correspondences	20
6.3	Transformation Estimation with RANSAC	20
6.4	Scoring and Verification	21
7	Pipeline Implementation	22
7.1	System Overview	22
7.2	Software Architecture	23
7.3	Implementation of Feature Extraction Methods	23
7.3.1	DISK-based Extraction	23
7.3.2	KeyNet-AffNet-HardNet Extraction	23
7.4	Feature Aggregation Implementation	24
7.5	Similarity Retrieval and Geometric Verification	24
8	Experimental Setup	25
9	Results	26
10	Discussion	27
11	Conclusion and Future Work	28
	References	29
	Abstract	33
	Sažetak	34

1 Introduction

The conservation and effective management of wildlife populations depend crucially on accurate methods for monitoring animal abundance and diversity. Traditional approaches, including manual identification, tagging are often invasive, costly, time-consuming and impractical at large scales or in remote environments. In recent years, significant progress has been achieved through the application of automated computer vision methods for wildlife monitoring, particularly though individual animal re-identification (Re-ID). Animal Re-ID, which refers to the task of identifying and matching individual animals across multiple observations solely on visual appearance, has emerged as a viable non-invasive method for ecological studies and conservation efforts.[1] Despite the advancements, current visual Re-ID systems still face substantial challenges, especially when applied to real-world scenarios characterized by significant appearance variations, environmental clutter, occlusions, and limited training data. Current methods typically require extensive manual labeling effort or species-specific model training, restricting their applicability in general ecological monitoring and conservation task so developing an efficient, flexible Re-ID framework capable of generalizing to virtually any animal species without requiring significant manual annotation or retraining represents an important and currently unmet need in conservation technology. The main aim of this thesis is just that, to develop a generalizable, robust animal Re-ID and population estimation system capable of performing reliably across diverse animal species with minimal manual intervention. Specifically, this thesis attempts to create a method suitable for real-world scenarios characterized by limited labeled data, eliminating the requirement for extensive manual labeling which can often be very difficult and error-prone cause it requires a certain expertise. To achieve these goals, the research was structured around two interlinked objectives:

1. **Evaluation and optimization of robust classification methodologies:** Before addressing the complex task of population estimation, it was essential to establish a strong foundation for animal Re-ID that generalizes effectively across species. Publicly available datasets, notably WildlifeReID[2], were leveraged for evaluation and testing. This phase involved extensive experimentation with robust feature extraction methods, including DISK[3] and LightGlue[4] combined with dimensionality reduction and aggregation approaches such as PCA, GMM and Fisher Vectors.
2. **Development of population size estimation method:** Following the establishment of a reliable and generalizable individual classification framework, the ultimate goal was population counting. Implementation of the Nested Importance Sampling (NIS)[5], a human-in-the-loop statistical framework leveraging pairwise similarity scores produced by the general-purpose Re-ID model. Critically, this method requires minimal human input and does not require additional training or extensive labeling for new animal species or datasets, thus significantly reducing manual labor and expertise requirements in practical scenarios. Several potential weak points were discovered in the paper itself which were attempted to address in this thesis.

To further validate the practicality and effectiveness of the developed methods in real-world conditions, original data was collected specifically for this thesis. Two trail cameras were deployed on Medvednica mountain for approximately five months, gathering data of local wildlife populations (mostly wild boars and deer). The final and crucial stage of this thesis involved applying the developed Re-ID and population estimation framework directly to this newly collected, unlabeled dataset.

2 Related Work

In this chapter, we'll review existing approaches to visual animal re-identification (Re-ID) with an emphasis on methods intended to generalize across species. The literature will be organized into thematic categories, from traditional computer vision techniques to modern deep learning methods, including hybrid methods that combine the two. We also discuss key challenges in the field, such as limited data, pose variation and open-set identification. We'll synthesize the findings from prior work and emphasize the gaps that this thesis is trying to address.

2.1 Traditional Computer Vision Approaches

Early animal Re-ID systems relied on hand-crafted visual feature and local descriptors. A prime example is HotSpotter[6], a SIFT-based[7] algorithm that identifies individuals by comparing local keypoints in images. HotSpotter uses viewpoint-invariant SIFT descriptors and scores matches by emphasizing the most distinctive keypoints (the titular "hot spots") on an animal's coat pattern. [8] compared SIFT, SURF and SuperPoint on identifying sunfish individuals and demonstrated that the number of matching keypoints between a pair of images is a strong indicator for the likelihood that they contain the same individual. These traditional local-feature methods do not require training on identity labels and are, by definition, species-agnostic, making them applicable to a wide variety of species. HotSpotter and similar algorithms have been successfully used on numerous species with distinctive markings, including zebras, giraffes, jaguars, ocelots and leopards.[9] The drawback, however, is that purely hand-crafted methods cannot fully exploit any available training data since their matching heuristic is fixed and may miss more subtle or global cues.

2.2 Species-Specific Trait-Based Re-Identification

Another line of research focuses on species-specific visual traits for re-identification. Instead of analyzing the entire image or generic local patches, these approaches isolate particular biometric features unique to the species at hand, for example the ear shapes of elephants, the outline of a shark’s dorsal fin, or the contour of a whale’s tail (fluke), and use them for re-identification. Both traditional image analysis and deep learning have been employed to recognize such traits. For example, algorithms such as CurvRank[10] and finFindR[11] compute and match the shapes of whale flukes or shark fins to identify individuals. These methods often don’t require individual identity labels for training, rather a detector or feature extractor for the trait can be trained using only that trait’s annotations, then match those features between images. The advantage of trait-specific methods is that they leverage highly distinctive features (when they are present) and can work with minimal ID labeling. However, a major limitation is that they are species-specific so each system is tailored for identifying only one species and usually it can’t be transferred to other species. This lack of generality is a significant barrier to wider use as it requires developing a new method for each studied species. This thesis is motivated by overcoming this limitation by developing a method, based on existing research, that does not depend on any single species’ unique trait.

2.3 Deep Learning-Based Methods

With rise of deep learning, a number of methods have been proposed that learn features directly from image data, without manual feature engineering. Modern deep learning approaches for animal re-ID typically employ CNNs (Convolutional Neural Networks) or related architectures to produce an embedding (feature vector) for each image, such that images of the same individual are close in feature space while those of different individuals are far apart. These can be trained either as classification (training a CNN to classify a fixed set of known individuals) or via metric learning (training the network to differentiate between same-individual vs. different individual image pairs). In general, metric learning approaches are favored for wildlife re-ID because they naturally handle open-set problems. A classification-based model treats re-ID as identifying one of N known individuals, which fails when a new individual outside the training set appears.

By contrast, metric-based methods learn a similarity function and can thus match new individuals by embedding comparison, without needing to retrain for new identities.[9]

2.4 Hybrid Approaches with Learned Local Features

A promising direction in animal re-id is to combine the strengths of traditional and deep learning methods into a hybrid pipeline. The idea is to use modern learned features from deep networks in place of the hand crafted ones, but still employ them in a local matching and aggregation framework as in traditional approaches. A good example of this is ALFRE-ID[9] pipeline (Aggregated Local Features for Re-Identification). ALFRE-ID uses a CNN extractor to extract local patch descriptors from an animal's coat pattern, then aggregates these local features into a global descriptor for the image.

2.5 Key Challenges and Limitations

Research in animal re-identification encounters several fundamental challenges that limit real-world performance. Key issues identified in the literature are:

TO DO: DOVRŠITI OVO PODPOGLAVLJE

3 Dataset and Preprocessing

Accurate and generalizable animal re-identification (Re-ID) system critically depend on the quality and diversity of datasets used for development and evaluation. This thesis leverages two distinct types of datasets: publicly available benchmark datasets[2] and original data collected specifically for validating the developed methods under realistic conditions, something that the available literature lacked. This chapter describes both categories in detail.

3.1 Public Benchmark Dataset: WildlifeReID-10k

WildlifeReID-10k[2] is a recently introduced large-scale benchmark explicitly curated for evaluating animal Re-ID methods across a variety of species. The primary goal of this dataset is to enable systematic benchmarking of general-purpose animal Re-ID methods.

3.1.1 Dataset Overview

WildlifeReID-10k is comprised of approximately 10,000 images, spanning 33 species. This is not a singular dataset, but rather an open-source collection of various Re-ID datasets from different projects and researches. The dataset has standardized splits into training, validation and testing subsets.

3.2 Original Dataset Collection: Medvednica

To validate the developed Re-ID and population estimation methods under realistic, uncontrolled conditions, original data collection was conducted as a part of this thesis. Two wildlife trail cameras were deployed on Medvednica on known animal trails. The data collection spanned approximately five months resulting in [NUM IMAGES]. It should

be noted that when population estimation is mentioned in this context, it refers only to identifying the number of unique individuals present in the collected images. To estimate the entire population of a species in an area would require placing more than two cameras in a systematic way and using statistical methods to account for a number of individuals that most likely never appear on the camera at all. This goes beyond the scope of this thesis but is an interesting topic for further research.

3.2.1 Dataset Collection Methodology

The two cameras were placed on animal trails, away from hiking trails to maximize the likelihood of animal activity. At the end of the data collection period, [PLACEHOLDER TOTAL IMAGES] were captured. These images mostly featured two target species prevalent in the region: wild boars (*sus scrofa*) and roe deer (*capreolus capreolus*).

3.2.2 Automatic Animal Detection and Classification

Given the practical constraints of manually reviewing and annotating thousands of captured images, an automated preprocessing pipeline was employed, leveraging existing state-of-the-art deep-learning models for animal detection and classification.

1. **Animal Detection:** The Megadetector[12] model, a deep learning-based object detector specifically trained for animal detection in wildlife camera trap images was used for automatic animal detection. Megadetector reliably localizes animals within images and outputs bounding boxes with high accuracy, significantly reducing manual effort by filtering out blank images (since a large number of images are false positives of cameras triggering due to tree branches moving in the field and similar) and producing tight bounding boxes that can be used in the pipeline.
2. **Species Classification:** Following detection, SpeciesNet[13], a convolutional neural network (CNN) model pre-trained to recognize multiple wildlife species, was utilized for automatic classification of detected animals into respective species categories. This was done to automate the separation of roe deer images and wild boar images.

3.2.3 Preprocessing Pipeline Overview

The complete preprocessing pipeline used in the creation of the original Medvednica dataset involved the following steps:

- Application of Megadetector to each collected image
- Cropping animal regions based on detected bounding boxes, resulting in animal-centered images ready for future feature extraction
- Automatic species classification of cropped animal images using SpeciesNet, separating species specific subsets (wild boars and roe deer)
- Filtering out low-confidence detections and classifications to maintain high dataset quality and consistency
- Manual pass through all of the images to correct any model mistakes

Through these preprocessing steps, the data collected on Medvednica was transformed into a structured, high-quality dataset of labeled (species level) wildlife images, ready to use in this thesis.

4 Feature Extraction

Feature extraction is a critical component of animal re-identification pipelines, converting raw image data into structured descriptors that capture distinctive visual characteristics. In this chapter I will be examining the theoretical principles which guided the choice of a method for detecting and extracting local features from images.

4.1 Local Feature Detection

Local features are small image patches, typically around corners, edges, or blobs that remain approximately stable under moderate changes in viewpoint, illumination and scale. Robust detection is critical in wildlife datasets, where pose, lighting and background vary dramatically from shot to shot. When we try to match two photos of the same animal, the first step is to pick out image patches that look the same even if the camera moved, zoomed, or the lighting changed. Those keypoints are what we call local features.

- **Corner detectors:** Harris corners [14] and the Shi–Tomasi update [15] look for pixels where the brightness changes strongly in two directions at once, basically sharp corners. They are quick and accurate, but on their own they do not handle large zoom differences so we have to repeat the test at several image scales.
- **Blob detectors:** Instead of sharp points, blob detectors look for spots that are brighter or darker than their surroundings at many sizes. The Laplacian-of-Gaussian (LoG) scale space [16] does this exactly while the Difference-of-Gaussians (DoG) used in SIFT [7] is a fast approximation of it giving it scale invariance.
- **Learning-based detectors:** Recent methods use convolutional neural networks to learn both where to place keypoints and how to describe them. SuperPoint [17],

R2D2 [18] and DISK [3] are three popular examples. By training on millions of image pairs, they learn to ignore lighting changes and background clutter, problems that often appear in camera-trap photos.

Choosing good keypoints boosts *recall* (more shared points between two images) and *precision* (fewer points that match the wrong place). That solid foundation lets later steps like feature aggregation and geometric verification work much better. In this paper, I opted to use learning-based detectors because they generally perform better than traditional ones like SIFT.

4.2 Descriptors

Once keypoints are detected, we compute descriptors that summarize the appearance of that local region. An effective descriptor should be discriminative (distinguishing one region from another) and robust (invariant to moderate geometric transformations).

- **Gradient-based descriptors:** Classic methods such as Scale-Invariant Feature Transform (SIFT) or Histogram of Oriented Gradients (HOG) capture local gradient orientations and magnitudes. They perform well in textured areas like stripes or distinctive fur patterns.
- **Learned descriptors:** Deep learning approaches train neural networks to produce feature vectors from image patches. HardNet [19], for example, is trained on many image pairs so that descriptors of matching patches are close together while mismatches are far apart, encoding subtle visual cues that hand-crafted methods might miss.
- **Affine normalization:** Methods such as AffNet estimate the local affine transform (stretch and skew) before computing the descriptor, so viewpoint changes affect the patch less.

In this thesis, I used two different methods for feature extraction, DISK and KeyNet+AffNet+HardNet.

4.3 DISK: Learning Local Features With Policy Gradient

DISK (DIScrete Keypoints) is an end-to-end trainable local feature framework that integrates detection and description into a single model. Unlike traditional pipelines with separate components, DISK uses a unified architecture trained with reinforcement learning to maximize correct feature matches. It outputs keypoint heatmap and a 128-D descriptor vector.

DISK operates by simultaneously generating:

- **Keypoint Heatmaps:** Dense probability maps indicating optimal keypoint locations.
- **128-dimensional Descriptors:** Robust, discriminative vectors characterizing local appearance.

The unified training ensures features are optimized specifically for reliable and accurate matching, even under challenging environmental conditions commonly observed in wildlife photography.

4.4 KeyNet, AffNet, and HardNet

The second approach employs a combination of three specialized deep neural networks, each addressing distinct aspects of robust feature extraction and matching:

4.4.1 KeyNet

KeyNet[20] focuses on identifying stable and repeatable keypoints. It utilizes convolutional neural network layers trained explicitly to detect points in images that are consistently recognizable under diverse transformations and viewing conditions. KeyNet significantly enhances keypoint stability, critical for reliable matching.

4.4.2 AffNet

AffNet complements KeyNet by predicting affine transformations to normalize identified keypoints. By correcting perspective distortions and other affine-related variations,

AffNet ensures that the extracted descriptors are viewpoint-invariant, thereby significantly improving matching accuracy across diverse animal poses and camera angles.

4.4.3 HardNet

HardNet computes robust descriptors optimized for matching tasks. Unlike traditional descriptor approaches (e.g., SIFT or SURF), HardNet employs a metric learning strategy that ensures descriptors corresponding to the same keypoint in different images are close in descriptor space, while descriptors from different keypoints are distinctively separated. This contrastive learning approach makes HardNet descriptors highly discriminative.

The integrated KeyNet+AffNet+HardNet extraction process involves:

1. **Image Preprocessing:** Images are converted to grayscale and standardized in size to ensure compatibility with the neural network inputs.
2. **Keypoint Detection and Affine Normalization:** KeyNet identifies keypoints, followed by AffNet's affine transformation predictions to adjust keypoint localizations for geometric consistency.
3. **Descriptor Computation:** Adjusted keypoints are passed to HardNet, which computes robust, contrastively-learned descriptors.
4. **Storage:** Resulting descriptors and keypoints are separately saved in structured HDF5 files, preserving clear and efficient data organization.

4.5 Storage Format and Management

Descriptors and keypoints are stored using the HDF5 file format, a hierarchical, binary storage format designed for managing large volumes of data. HDF5 ensures rapid read-write access, efficient compression to reduce storage demands, and structured data management.

In summary, feature extraction methods discussed in this chapter are the foundation for reliable and robust animal re-identification in this thesis, ensuring that subsequent

analysis and matching procedures benefit from highly descriptive and discriminative image representations.

5 Feature Aggregation and Representation

This chapter describes the process of aggregating local feature descriptors extracted from animal images into compact global representations suitable for efficient comparison and re-identification. As described in [9] Fisher Vectors (FV) were utilized, built on PCA and Gaussian Mixture Models (GMMs).

5.1 Training PCA and GMM models

Feature descriptors extracted from images can be high-dimensional and noisy. Thus, dimensionality reduction and normalization are performed to improve efficiency and accuracy.

Initially PCA is applied to decorrelate and reduce the dimensionality of the descriptors. PCA projects the original descriptors onto a lower-dimensional subspace defined by the principal eigenvectors of the covariance matrix computed from the descriptors. This is also important later on for the computation of fisher vectors because they produce very large descriptors.[9]

After dimensionality reduction, a Gaussian Mixture Model (GMM) is trained on the reduced descriptors. A GMM models the descriptor distribution as a weighted sum of K Gaussian components, each parameterized by mean vectors $\{\mu_k\}$, covariance matrices $\{\Sigma_k\}$, and weights $\{\pi_k\}$. Specifically, a GMM is defined as:

$$p(x) = \sum_{k=1}^K \pi_k \mathcal{N}(x|\mu_k, \Sigma_k), \quad (5.1)$$

where $\mathcal{N}(x|\mu_k, \Sigma_k)$ denotes the probability density function of the k -th Gaussian com-

ponent. This codebook (the trained GMM) captures the visual vocabulary and statistical distribution of the reduced descriptors from the training images.[9]

In simpler terms, PCA finds the directions in which the descriptor values vary the most and keeps only those, so each feature becomes a shorter, uncorrelated vector that is faster to store and compare. Then we model how these compact vectors are distributed. GMM model assumes that the whole set of vectors can be grouped into a number of smooth clusters, each cluster described by it's mean, spread and relative weight in the mixture. Together, these clusters form a codebook meaning that every new descriptor can be represented as a weighted combination of the nearest clusters, giving us both a compressed encoding and a way to compute similarity for matching or classification which is of crucial importance in re-identification task.

5.2 Fisher Vector Computation

Fisher vectors (FV) are an image representation technique that captures statistical deviations of local features from a visual vocabulary (codebook) modeled by GMM. FV measures how features differ from typical patterns in a reference GMM and they encode both first-order differences (mean shifts) and second-order differences (variance changes) in feature distribution. The output is a high-dimensional vector that acts as a "signature" of an image's unique feature distribution. Basically, given a set of descriptors extracted from a single image, FV encodes how these descriptors deviate from the trained GMM parameters. Fisher Vectors represent the gradients of the log-likelihood function with respect to the GMM parameters. Formally, given an image with descriptors $X = \{x_t\}_{t=1}^T$ where λ is a vector of GMM parameters and u_λ is a probability density function modeling the distribution of the data.

This gradient is formally known as the score function[9]. It measures the sensitivity of the log-likelihood to changes in the model parameters, essentially indicating how to adjust the GMM to better fit the observed features.

$$G_\lambda^X = L_\lambda \log u_\lambda(X), \quad (5.2)$$

When constructing a FV for an image, a set of local features is assumed to be independent which means that the final descriptor can be constructed as a sum of Fisher Vectors for each local feature:[9]

$$G_{\lambda}^X = \sum_{t=1}^T L_{\lambda} \nabla_{\lambda} \log u_{\lambda}(x_t) \quad (5.3)$$

The matrix L_{λ} is derived from the Fisher Information Matrix and is used to "whiten" or to normalize the gradients[21], making them more discriminative and suitable for use in classification later on.

Following the standard practice, this vector is computed separately for means and variances, resulting in a final vector of dimension $2KD$, where K is the number of Gaussian components and D the dimensionality of the PCA-reduced descriptors. L2 and power normalization have also been applied since they have been shown[22] to generally improve the performance of the method.

FV have also been chosen because they have been shown to perform better than bag-of-words approaches.[23]

5.3 Parameter Choices and Rationale

Several key parameters influenced the effectiveness of Fisher Vectors for the re-identification task during the development:

- **Number of PCA components (N_{PCA}):** A balance between dimensionality reduction and retention of discriminative information.
- **Number of GMM components (N_{GMM}):** Controls the granularity of the visual codebook. More components typically yield richer representations but increase computational complexity.
- **Maximum descriptors sampled ($MAX_GMM_DESCRIPTORS$):** Limits the number of descriptors sampled for training PCA and GMM to manage memory usage and computational load.

These parameters were empirically determined based on experimenting and computational constraints. Specifically, the PCA dimension was selected to retain approximately 95% of descriptor variance. The number of GMM components was chosen to balance accuracy and computational efficiency. The best results were achieved, with 96 PCA components, 256 GMM components and 200,000 descriptors. The larger number of maximal components could perhaps prove better results but computational constraints were a limiting factor.

6 Geometric Verification

Geometric verification is used to make sure that matched keypoints between images truly correspond to the same physical locations on an animal reducing false positives that arise from visual similarities alone. This step is crucial in refining the initial matches obtained through descriptor-based feature matching by enforcing spatial consistency.

6.1 Theoretical Foundation

Local feature descriptors such as DISK[3] or KeyNet combined with AffNet and HardNet [20, 24, 19], offer robust initial matches based on visual similarity. However, matches that are generated purely from descriptor similarity may include mistakes, especially in visually cluttered or repetitive environments common in wildlife images. Geometric verification, as suggested in [9], addresses this by validating matches through spatial coherence checks.

6.2 Establishing Initial Correspondences

Initial correspondences are established through descriptor matching using LightGlue[4], an attention-based method. Initially methods like Lowe’s ratio test were considered but matching with LightGlue proved significantly faster and more accurate, effectively eliminating the need for traditional filtering methods.

6.3 Transformation Estimation with RANSAC

The core of geometric verification involves estimating a homography transformation between matched keypoints using RANSAC (Random Sample Consensus). RANSAC iteratively samples minimal subsets of correspondences, computes candidate transfor-

mations and evaluates their agreement with the remaining matches. Matches that fall within a defined spatial threshold are considered inliers.

6.4 Scoring and Verification

Matches are assessed primarily by their inlier count, the number of matches consistent with the estimated homography, and their spatial distribution:

- **Inlier count:** Directly quantifies geometric consistency, higher counts typically indicate reliable matches
- **Inlier ration:** Provides normalization ... (dovršiti ovu rečenicu)
- **Combined scoring:** Integrates geometric consistency with descriptor similarity (fisher vector distances) to leverage both spatial and appearance information.

Penalties for low inlier counts or poor geometric coherence discourage accepting borderline matches, improving the overall system reliability and accuracy.

7 Pipeline Implementation

This chapter details the practical implementation of the entire animal re-identification pipeline, connecting feature extraction, feature aggregation, geometric verification, and the classification components into a complete end-to-end workflow. It provides insight into the project's architecture, optimization strategies and computational considerations.

7.1 System Overview

The implemented pipeline consist of sequential, interconnected stages:

1. **Image acquisition and pre-processing**
2. **Segmentation**
3. **Local feature extraction and description**
4. **Feature aggregation (Fisher Vector computation)**
5. **Similarity retrieval and geometric verification**

The theory most of this we covered in the preceeding chapters so here we'll go in-depth about how each component was implemented. The architecture itself was mostly inspired by the ALFRE-ID system describerd in [9] with some modifications and additions in an effort to improve on it.

Figure ?? Za ovdje napraviti sliku koja vizualizira cijeli pipeline.

7.2 Software Architecture

The implementation is done entirely in Python, leveraging popular libraries for efficiency and reproducibility:

- **PyTorch** for deep-learning based methods (local feature extraction).
- **OpenCV** for image processing and pre-processing tasks.
- **scikit-learn** for PCA and GMM training.
- **NumPy and SciPy** for numerical computations.
- **h5py** for efficient storage and retrieval of large descriptor datasets.

The codebase is modular, enabling easy experimentation and component swapping.

7.3 Implementation of Feature Extraction Methods

Two (? Ovo možda naraste na 3) primary feature extraction pipelines were implemented:

7.3.1 DISK-based Extraction

1. Load the pre-trained DISK model [3].
2. Images are resized to 640×640 pixels and converted to grayscale.
3. Keypoints and descriptors are directly computed by a single forward pass.
4. Store resulting descriptors and keypoints efficiently.

7.3.2 KeyNet-AffNet-HardNet Extraction

Instead of DISK, a combination of KeyNet-AffNet-HardNet can be used for feature extraction, this method, as we will discuss in the later chapters proved to deliver slightly better results than DISK on most datasets tested, but was also slower and more memory intensive.

1. Extract initial keypoints with KeyNet [25].

2. AffNet[24] performs local affine normalization of detected patches.
3. Generate descriptors from normalized patches using HardNet [19].
4. Descriptors and keypoints are stored separately in HDF5 files for downstream tasks.

For memory purposes, the number of keypoints is capped at 5000, this number was selected as a tradeoff between speed, accuracy and memory limitations.

7.4 Feature Aggregation Implementation

Feature descriptors are aggregated using PCA and GMM-based Fisher Vector representation (described previously in Chapter 5.2). Practical considerations in the implementation include:

1. **Descriptor Sampling:** Random sampling of descriptors to manage computational complexity.
2. **PCA Training:** PCA [26] reduces dimensionality and applies whitening.
3. **GMM Training:** A Gaussian Mixture Model (GMM) [26] with diagonal covariance is fitted to PCA-reduced descriptors.
4. **Fisher Vector Computation:** Fisher vectors encode gradients of log-likelihoods relative to trained GMM parameters.

7.5 Similarity Retrieval and Geometric Verification

The retrieval component involves two key stages:

1. **Initial retrieval:** Cosine similarity between Fisher Vectors rapidly identifies potential matches.
2. **Geometric Verification:** Matches are further verified using RANSAC-based homography estimation (explained in detail in chapter XYZ)

The geometric verification implementation includes optimizations such as descriptor normalization and coordinate normalization to enhance RANSAC performance.

8 Experimental Setup

- Hardware
- Evaluacijske metrike
- Evaluacijske metrike za prebrojavanje individualaca

9 Results

- Rezultati klasifikacije (top-1, top-n acc), rezultati prebrojavanja individualnih životinja, rezultati prebrojavanja na mom datasetu
- Vizualizacije, tablice
- Usporedba utjecaja hiperparametara testiranih

10 Discussion

- Interpretation of results.
- Successes and limitations.
- Comparison with related work.

11 Conclusion and Future Work

- Summary of contributions and findings.
- Potential extensions and open questions.

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Abstract

Deep learning-based animal re-identification for non-invasive wildlife monitoring and conservation

Matej Marić

Reliable methods for monitoring wildlife populations are essential for effective conservation and ecological management. This thesis addresses the challenge of developing an automated visual system capable of identifying individual animals and estimating their population size across various species, particularly when limited labeled data is available. The primary goal was to establish a practical approach that can generalize effectively without the need for extensive manual annotation or species-specific model retraining. The developed methods were initially validated on publicly available benchmark datasets and subsequently tested using original data collected through trail cameras placed on Medvednica mountain in Zagreb, Croatia. The research specifically aimed at evaluating the system's practical applicability and accuracy in real-world conditions, focusing on local populations of wild boars (*Sus scrofa*) and roe deer (*Capreolus capreolus*).

Keywords: Animal Re-identification; Wildlife Monitoring; Population Estimation; Computer Vision; Conservation

Sažetak

Duboko učenje za ponovnu identifikaciju životinja u neinvazivnom praćenju divljih vrsta i očuvanju prirode

Matej Marić

Pouz dane metode praćenja populacija divljih životinja ključne su za učinkovito očuvanje i ekološko upravljanje. Ovaj rad bavi se izazovom razvoja automatiziranog vizualnog sustava sposobnog za identificiranje pojedinačnih životinja i procjenu veličine njihove populacije među različitim vrstama, posebno kada je dostupa ograničen broj označenih podataka. Primarni cilj bio je uspostaviti praktičan pristup koji se može učinkovito generalizirati bez potrebe za opsežnim ručnim označavanjem ili ponovnim treniranjem modela specifičnih za vrstu. Razvijene metode u početku su validirane na javno dostupnim skupovima podataka, a potom testirane korištenjem originalnih podataka prikupljenih putem kamera na stazama postavljenih na Medvednici u Zagrebu, Hrvatska. Istraživanje je posebno bilo usmjereno na procjenu praktične primjenjivosti i točnosti sustava u stvarnim uvjetima, s naglaskom na lokalne populacije divljih svinja (*Sus scrofa*) i srna (*Capreolus capreolus*).

Ključne riječi: Re-identifikacija životinja; Nadzor Divljači; Procjena Populacije; Računalni Vid; Očuvanje Prirode